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Physics-informed neural networks for coupled rate-and-state friction and pore-pressure evolution

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SUMMARY

Earthquake fault slip arises from nonlinear coupling among frictional evolution, elastic loading and pore-pressure changes. When pore pressure evolves dynamically, the resulting hydromechanical rate-and-state models can be stiff and strongly coupled, making parameter inversion computationally demanding. Here we develop a physics-informed neural network (PINN) solver for a coupled spring-slider system that combines rate-and-state friction with pore-pressure/porosity evolution. The network approximates the time-dependent state variables and is trained by enforcing the governing differential equations together with initial conditions and, for inverse problems, observational constraints. To improve training stability, we employ adaptive inverse-residual weighting and a two-stage optimization schedule, in which Adam is followed by limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) optimization. In forward simulations, PINN predictions closely match a Runge–Kutta reference solution across steady sliding and slow-slip transients, with normalized mean squared error below 0.08 and Pearson correlation coefficient above 0.975 for block velocity and frictional shear stress in the cases tested. In inverse experiments, the framework recovers the applied normal stress from noisy shear-stress observations; uncertainty increases with noise amplitude, but the ensemble mean remains stable, and at the highest noise level considered ($q = 1$) the inferred normal stress deviates by less than ∼1 per cent from the reference value. These results suggest that PINNs provide a differentiable alternative for forward modelling and parameter inversion in coupled hydromechanical rate-and-state fault models.

Key words: Friction; Inverse theory; Machine learning.

1 INTRODUCTION

Rate-and-state friction (RSF) provides a constitutive framework for fault slip in which friction depends on sliding velocity and a state variable that evolves with contact history (J.H. Dieterich 1979; A. Ruina 1983). RSF has been widely used to study fault stability and the emergence of diverse slip styles, from stable creep to slow-slip events and dynamic rupture, across laboratory and crustal scales (Marone, 1998; D.M. Veedu & S. Barbot 2016; Y. Luo & J.-P. Ampuero 2018; C. Mei & W. Wu 2021; J. Wang *et al.* 2025). RSF has been applied from laboratory-scale fault sliding to natural earthquake-scale problems, providing a theoretical foundation for understanding seismic activity and the long-term evolution of fault history (S. Barbot *et al.* 2012; Dal Zilio, Lapusta & Avouac 2020). Furthermore, because fault strength depends on effective normal stress, fluid pressure variations can strongly modulate RSF-driven instabilities. Infiltration and diffusion of pore pressure along a fault can destabilize sliding by reducing effective normal stress, as shown in seminal analyses of coupled slip and pore-pressure diffusion (P. Segall & J.R. Rice 1995) and supported by laboratory friction experiments under elevated pore pressure. In natural subduction systems, pore-fluid processes are widely regarded as a first-order control on slow slip behaviour (D.M. Saffer & H.J. Tobin 2011). Transient pore-pressure increases may promote slow slip by reducing effective stress, whereas

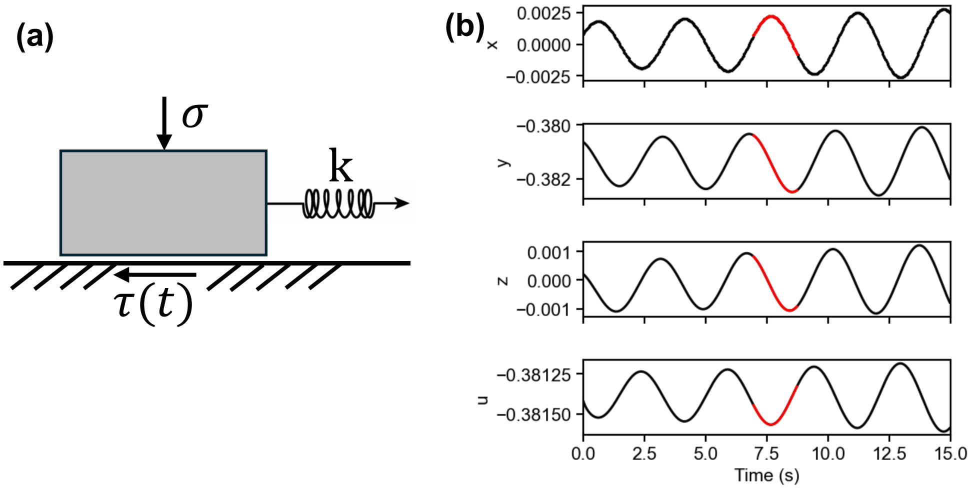

Figure 1. (a) Simple spring slider model. (b) Temporal evolution of $\zeta_2 = [x, y, z, u]$, solved by Fourth-Order Runge–Kutta method (RK4). The highlight curve represents the portion used for the PINN solution in Section 3.2.

dilatant strengthening may stabilize slip through slip-induced porosity increase, local pore-pressure reduction and the resulting recovery of effective normal stress (P. Segall *et al.* 2010). The interplay among fluid overpressure, pore-pressure diffusion and dilatancy thus provides a physically self-consistent framework for explaining the initiation, evolution and termination of slow slip events (SSEs).

Incorporating pore-pressure evolution into RSF-based models is therefore essential for capturing hydromechanical feedbacks that may govern transitions between aseismic and seismic slip. However, fully coupled formulations introduce additional state variables and feedbacks that are stiff and strongly nonlinear, especially when diffusion and dilatancy/compaction evolve on timescales comparable to slip. Standard numerical solvers can reproduce these dynamics (R. Tang *et al.* 2025), but they require careful discretization and time stepping and parameter estimation becomes costly when repeated forward simulations are needed for inversion.

Physics-Informed Neural Networks (PINNs) offer a complementary approach for such forward-and-inverse problems governed by differential equations. PINNs train neural networks to satisfy the governing equations by embedding the residuals of the differential system (and associated initial/boundary conditions) directly in the loss function, optionally together with observational constraints (M. Raissi *et al.* 2019). In geophysics, PINNs have been applied within RSF settings for efficient simulation and parameter inference: for example, modelling slow-slip in simplified spring-slider systems (R. Fukushima *et al.* 2023), inferring RSF parameters using acoustic-emission information (P. Borate *et al.* 2024) and extending to elastic-dynamics and data-assimilation settings (C. Rucker & B. A. Erickson 2024; R. Fukushima *et al.* 2025). Yet, most PINN–RSF studies still assume prescribed pore-pressure histories or neglect pore-pressure evolution, despite the central role of diffusion-driven pressure changes in controlling fault stability. Extending PINNs to coupled hydromechanical RSF models is non-trivial because stiffness and strong coupling can degrade training stability and identifiability in inverse settings.

Here, we develop a PINN for a coupled hydromechanical spring-slider model that accounts simultaneously for rate-and-state friction and pore-pressure/porosity evolution through a non-dimensionalized ordinary differential equation (ODE) system for slip velocity, shear stress, state, porosity and pore fluid pressure (A. Gualandi *et al.* 2023). We use the same framework for (i) forward simulation across steady sliding and slow-slip regimes and (ii) inverse estimation of key frictional and hydraulic parameters from sparse and noisy observations. To improve convergence in this coupled setting, we adopt adaptive loss weighting and multistage optimization strategies. The paper is structured as follows: Section 2 introduces the governing spring-slider model, the PINN formulation and the experimental design. Section 3 presents forward and inverse results. Section 4 evaluates performance across slip regimes, robustness to noise and the impact of optimization strategies for multicoupled systems. Section 5 concludes.

2 DATA AND METHODOLOGY

2.1 Rate-and-state friction

To model the coupled hydromechanical dynamics of the fault, we employ the classical spring-slider system (Fig. 1a). This computational setup, composed of a slider block connected by a spring to a rigid boundary, is widely used to represent the elastic loading imposed by the surrounding lithosphere on the fault (J.H. Dieterich 1979). Given that the PINN approach is designed to learn approximate solutions directly from the governing ordinary differential equations (ODEs), our primary methodological step is to first establish the complete set of coupled ODEs that define this spring-slider system, specifically incorporating the evolution of pore pressure. We adopt the classical rate-and-state friction law motivated by laboratory experiments (J.H. Dieterich 1979; A. Ruina 1983; C. Marone *et al.* 1990). RSF has shown its modelling capacity from laboratory scales to natural earthquake scales (J.P. Ampuero & A.M. Rubin 2008).

We adopt the most basic and commonly used form, where the shear stress τ_f of a fault is described as J.H. Dieterich (1979)

$$\tau_\mathrm{f} = \sigma_\mathrm{n}\mu = (\sigma_{\mathrm{n}0} - p)\left[\mu_0 + a\ln\left(\frac{v}{v_*}\right) + b\ln\left(\frac{v_*\theta}{L}\right)\right], \tag{1}$$

where the effective normal stress σ_n is the difference between total normal stress $\sigma_{\mathrm{n}0}$ and pore fluid pressure p, a is the direct effect parameter, b is the evolution effect parameter, μ_0 and v_* are reference values of the friction coefficient and sliding velocity, L is a critical slip distance required to evolve into a new state in response to a velocity perturbation and θ is the state variable often interpreted as the average lifetime of grain contacts. Here we adopt the so-called 'slip law', which leads to an exponential relaxation of the friction force after an abrupt velocity change (A. Ruina 1983):

$$\dot{\theta} = -\frac{v\theta}{L}\ln\left(\frac{v\theta}{L}\right). \tag{2}$$

The loading shear stress τ_l is given by

$$\tau_\mathrm{l} = k(v_* t - \delta), \tag{3}$$

where k is the elastic stiffness of the spring, v_* is the long-term loading velocity of the driving plate, t is time and δ is the amount of slip underwent by the block. Complete negligence of inertial effects leads to situations in which the slip velocity increases without bound, making simulations of repeated stick-slip cycles impossible. We use the radiation-damping approximation in place of explicit inertia to suppress unbounded acceleration and maintain stable simulations of repeated stick-slip cycles. In the Fourier-domain representation of the elastodynamic kernel, the radiation-damping term arises as the zeroth-order term in the high-frequency asymptotic expansion of the impedance for $|\omega| \gg kc$, which yields the local relation

$$\tau_\mathrm{v} = \eta v = \tau_\mathrm{l} - \tau_\mathrm{f}, \tag{4}$$

where $\eta = \frac{G}{2C_\mathrm{s}}$ is the radiation damping coefficient, G is the shear modulus and C_s is the shear waves speed. The damping term is introduced to represent energy dissipation due to seismic wave radiation in a continuum.

Based on experiments by C. Marone et al. (1990) on the shearing of gouge layers of quartz sand at an effective confining stress of 150 MPa, P. Segall and J.R. Rice (1995) proposed the following expression for the rate of change of porosity ϕ

$$\frac{\mathrm{d}p}{\mathrm{d}t} = c^*(p_\infty - p) - \frac{\dot{\phi}}{\beta}, \tag{5}$$

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = -\frac{v}{L_\phi}\left(\phi - \phi_0 - \epsilon\ln\left(\frac{v}{v_*}\right)\right), \tag{6}$$

where c^* is the diffusivity, p_∞ is the pore pressure in the surrounding of the fault, β is the product of the elastic component of the porosity and the combined compressibility of the fluid in the pores and the elastic pores, ϕ_0 is a reference plastic porosity, L_ϕ is a characteristic relaxation distance and ϵ is a dilatancy coefficient. We define the state vector ζ_1, which characterizes the system's state, as consisting of four variables: $\zeta_1 = [v, \tau_\mathrm{f}, \phi, p]$ (Fig. 1b).

Based on eqs (1) and (4), the derivatives of v and τ_f can be obtained as follows

$$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{k}{\eta}(v_* - v) - \frac{1}{\eta}\frac{\mathrm{d}\tau_f}{\mathrm{d}t}, \tag{7}$$

$$\frac{\mathrm{d}\tau_\mathrm{f}}{\mathrm{d}t} = \frac{\mathrm{d}\sigma_\mathrm{n}}{\mathrm{d}t}\frac{\tau_\mathrm{f}}{\sigma_\mathrm{n}} + a\sigma_\mathrm{n}\left[\frac{1}{v}\frac{\mathrm{d}v}{\mathrm{d}t} + \beta_1\frac{1}{\theta}\frac{\mathrm{d}\theta}{\mathrm{d}t}\right], \tag{8}$$

where $\beta_1 = \frac{b}{a}$. To facilitate the construction and enforcement of the residual loss in the PINN framework, eqs (7) and (8) can be further written as:

$$\frac{\mathrm{d}v}{\mathrm{d}t} = v\left\{\frac{1}{a\sigma_{\mathrm{n}0}}\frac{\mathrm{d}\tau_f}{\mathrm{d}t}\frac{1}{1-\frac{p}{\sigma_{\mathrm{n}0}}} + \frac{1}{a\sigma_{\mathrm{n}0}}\frac{\mathrm{d}p}{\mathrm{d}t}\frac{1}{1-\frac{p}{\sigma_{\mathrm{n}0}}}\frac{\tau_\mathrm{f}}{\sigma_{\mathrm{n}0}}\frac{1}{1-\frac{p}{\sigma_{\mathrm{n}0}}}\right.$$
$$\left. + \frac{v}{L}\left[(\beta_1 - 1)\ln\left(\frac{v}{v_*}\right) + \frac{1}{1-\frac{p}{\sigma_{\mathrm{n}0}}}\frac{\tau_\mathrm{f} - \mu_0\sigma_{\mathrm{n}0}}{a\sigma_{\mathrm{n}0}} + \frac{\mu_0}{a}\frac{p/\sigma_{\mathrm{n}0}}{1-\frac{p}{\sigma_{\mathrm{n}0}}}\right]\right\}. \tag{9}$$

$$\frac{\mathrm{d}\tau_f}{\mathrm{d}t} = k(v_* - v) - \eta\frac{\mathrm{d}v}{\mathrm{d}t} \tag{10}$$

Based on eqs (5), (6), (9) and (10), we can simplify and summarize the model with the following variables (A. Gualandi *et al.* 2023)

$$T = \tfrac{v_* t}{L}, \tag{11a}$$

$$x = \ln\left(\tfrac{v}{v_*}\right), \tag{11b}$$

$$y = \tfrac{\tau_f - \tau_0}{a\sigma_{\mathrm{n}0}} \tag{11c}$$

$$z = \tfrac{\phi - \phi_0}{\lambda\beta\sigma_{\mathrm{n}0}}, \tag{11d}$$

Table 1. Model parameters used in the rate-and-state friction formulation.

Symbol	Description	Value
v_*	Loading velocity	10×10^{-6} m s^{-1}
k	Spring stiffness	14.8×10^{9} Pa m^{-1}
a	Direct effect parameter	0.01
b	Evolution effect parameter	0.012
β_1	Ratio b/a	1.2
L	Critical slip distance	3×10^{-6} m
L_ϕ	Porosity evolution length scale	3×10^{-7} m
μ_0	Reference friction coefficient	0.64
G	Shear modulus	31×10^{9} Pa
c_s	Shear-wave speed	3420.25 m s^{-1}
p_∞	Background pore pressure	$1.013\,25 \times 10^{5}$ Pa
ϕ_0	Reference porosity	0.075
β	Effective compressibility	7.575×10^{-11} Pa^{-1}
ϵ	Dilatancy coefficient	-0.017×10^{-3}
c^*	Hydraulic diffusivity	10 s^{-1}
σ_{n0}	Applied normal stress	17.003×10^{6} Pa
x_{init}	Initial logarithmic dimensionless slip velocity	1.485×10^{-3}
y_{init}	Initial dimensionless shear stress change	-3.807×10^{-1}
z_{init}	Initial dimensionless porosity change	1.626×10^{-4}
u_{init}	Initial dimensionless pore-pressure	-3.814×10^{-1}

$$u = -\frac{p}{\lambda\sigma_{n0}}, \tag{11e}$$

where $\tau_0 = \mu_0\sigma_{n0}$ and $\lambda = \frac{a}{\mu_0}$.

In this way, the time derivative can be converted into the non-dimensional time derivative via $\frac{\mathrm{d}}{\mathrm{d}t} = \frac{v_*}{L}\frac{\mathrm{d}}{\mathrm{d}T} = \frac{v}{L}\dot{}$

$$\dot{x} = \frac{e^x[(\beta_1-1)x(1+\lambda u)+y-u]+\kappa\left(\frac{v_*}{v_*}-e^x\right)-\dot{u}\frac{(1+\lambda y)}{(1+\lambda u)}}{1+\lambda u+\nu e^x}, \tag{12a}$$

$$\dot{y} = \kappa\left(\frac{v_*}{v_*} - e^x\right) - \nu e^x\dot{x}, \tag{12b}$$

$$\dot{z} = -\rho e^x\left(\beta_2 x + z\right), \tag{12c}$$

$$\dot{u} = -\alpha - \gamma u + \dot{z}, \tag{12d}$$

where we have introduced the following parameters to simplify the notation:

$$\kappa = \frac{kL}{a\sigma_{n0}}, \tag{13a}$$

$$\nu = \frac{\eta v_*}{a\sigma_{n0}}, \tag{13b}$$

$$\rho = \frac{L}{L_\phi}, \tag{13c}$$

$$\beta_2 = -\frac{\epsilon}{\lambda\beta\sigma_{n0}}, \tag{13d}$$

$$\alpha = \frac{Lc^* p_\infty}{v_*\lambda\sigma_{n0}}, \tag{13e}$$

$$\gamma = \frac{Lc^*}{v_*}. \tag{13f}$$

Eq. (13) defines a set of non-dimensional control parameters for the coupled fault-fluid system. Specifically, κ characterizes the relative stiffness of the loading system, ν the normalized damping strength, ρ the relative rate of porosity evolution compared with friction-state evolution, β_2 the strength of dilatancy coupling, α the background pore-pressure forcing and γ the hydraulic relaxation rate. Together, these parameters determine how elastic loading, friction, porosity evolution and pore-pressure diffusion interact to control fault stability and slow-slip behaviour.

Finally, we obtain a new state vector that fully characterizes the system's state, composed of dimensionless variables $\zeta_2 = [x, y, z, u]$ (Fig. 1b), and their corresponding derivatives $\dot{\zeta}_2 = [\dot{x}, \dot{y}, \dot{z}, \dot{u}]$. The definitions and numerical values of all parameters in Section 3 are provided in Table 1. The formula presented here follows from the derivation in A. Gualandi *et al.* (2023). To justify our parameter choices, we ran additional simulations to test how the slip behaviour responds to ϵ across hydraulic diffusion regimes. At low c^*, the fault zone retains pore-pressure perturbations, so changes in ϵ strongly affect the slip evolution. At high c^*, hydraulic diffusion rapidly relaxes these perturbations, so the effect of ϵ becomes much weaker (Fig. S1, Supplementary Material). These scalings show that our parameter choices span the transition from a nearly undrained regime to a more efficiently drained regime.

2.2 Neural network modelling

In this study, we follow the original framework of PINN (M. Raissi *et al.* 2019) to solve for the state vector ζ_2 in the RSF problem, where the parameters of the neural network serve as the solution tool. The network's input is a 1-D time variable t and the output is the state vector ζ_2. The specific architecture of the neural network is implemented using the PyTorch framework (A. Paszke *et al.* 2019),

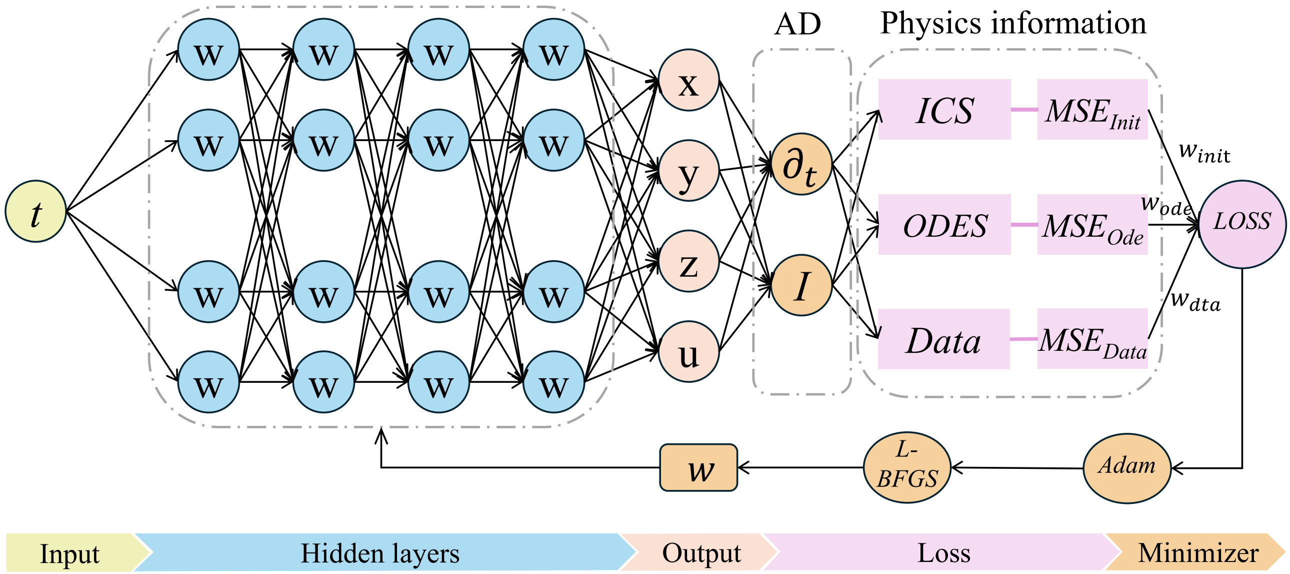

Figure 2. The architecture of the physics-informed neural networks.

configured as follows (Fig. 2): The input layer consists of a single neuron that receives the time variable t; four hidden layers follow, each containing 200 neurons and employing the hyperbolic tangent function (tanh) as the activation function to ensure nonlinear mapping capability. The number of neurons was treated as a hyperparameter that affects the representational capacity of the PINN. We tested several network widths using the same loss terms and optimization strategy, and found that 200 neurons provided a stable solution with a good balance between fitting accuracy and computational cost (Fig. S2, Supplementary Material). In a feedforward network, each layer feeds the next layer through a nested transformation

$$o_l(o_{l-1}) = \tanh(W_l o_{l-1} + b_l), \tag{14}$$

where for each layer $l = 1, \ldots, L$, $W_l \in \mathbb{R}^{m_l \times m_{l-1}}$ is the weight matrix, and $b_l \in \mathbb{R}^{m_l}$ is the bias vector. The output layer consists of four neurons corresponding to the components of the state vector ζ_2.

To incorporate physical constraints, the model utilizes PyTorch's built-in automatic differentiation (AD) functionality. AD precisely computes the derivatives of the state vector $\dot{\zeta}_2 = [\dot{x}, \dot{y}, \dot{z}, \dot{u}]$, based on the computation graph, thus avoiding the discretization errors of numerical differentiation inherent in traditional methods and makes embedding complex ODEs into the loss function possible.

2.3 Loss function

PINN approximates the mapping between temporal points and the solution of the ODE by training a neural network to fit the state vector ζ_2 at sampling points. The neural network learns the behaviour of the equation through the definition of a loss function.

For the forward problem, the total loss function L_{total} typically consists of two parts: one reflecting the initial conditions in the time domain, and the other representing the ODE residuals at the sampled points in the domain. The loss function L_{total} is defined as

$$L_{\text{total}} = w_{\text{ini}} L_{\text{ini}} + w_{\text{ode}} L_{\text{ode}}, \tag{15}$$

where L_{ini} represents the loss related to the initial conditions, and L_{ode} represents the residual of the ODE at the sampled points in the domain. w_{ini} and w_{ode} denote the weights of these two losses. Here, we set $w_{\text{ini}} = 0.5$ and $w_{\text{ode}} = 1$. Because the initial conditions loss only involves the four variables of ζ_2 at the initial time point $[x_0, y_0, z_0, u_0]$, meaning it is limited to a single time point, which is typically easier to satisfy. On the other hand, the ODE residual L_{ode} involves multiple sampled points over the entire time domain, making it more complex and harder to optimize. By assigning a higher weight to the ODE residual in the loss function, we emphasize the model's priority to ensure that the solution adheres to the constraints of the ODE throughout the entire time domain.

The loss of the mean squared error of the initial condition is given as

$$L_{\text{ini}} = \text{MSE}_{\text{ini}} = \frac{1}{4} \sum_{i \in \{x,y,z,u\}} \| i_{\text{NN}}(0) - i_{\text{ini}} \|_2^2 , \tag{16}$$

where $i_{\rm NN}$ is the output of the neural networks and $i_{\rm ini}$ is the given initial condition at $t = 0$. The specific values can be found in Table 1. The mean squared error resulting from the residuals of the ODE equations is defined as

$$L_{\rm ode} = {\rm MSE}_{\rm ode} = \frac{1}{4}\sum_{i\in\{x,y,z,u\}}\left(w_i\sum_{j=1}^{N}\left\|\frac{{\rm d}i_{\rm NN}(t_j)}{{\rm d}t} - \frac{{\rm d}i(t_j)}{{\rm d}t}\right\|_2^2\right), \tag{17}$$

where N represents the number of sampled points, and w_i denotes the residual weights of the four variables in ζ_2. Since the magnitudes of the four variables differ in ζ_2, we use an inverse residual weighting mechanism to balance the contribution of different physical equations to the ODE loss. This mechanism dynamically adjusts the relative importance of each residual term, ensuring a balanced contribution from each physical equation to the overall ODE loss. The residual weighting is calculated using the mean absolute residual for each equation, defined as follows

$$\mu_i = \frac{1}{N}\sum_{j=1}^{N}\left\|\frac{{\rm d}i_{\rm NN}(t_j)}{{\rm d}t} - \frac{{\rm d}i(t_j)}{{\rm d}t}\right\|_2, \quad i \in \{x, y, z, u\} \tag{18}$$

The weights are clamped to ensure they remain within a specified range [0.1, 10.0]:

$$w_i = \max\left(0.1, \min\left(10.0, \frac{1}{\mu_i}\right)\right). \tag{19}$$

For the inverse problem, the total loss function $L_{\rm total}$ typically consists of three parts. In addition to $L_{\rm ini}$ and $L_{\rm ode}$ in the forward problem, it also includes the data residual loss $L_{\rm data}$ at the sampled data points. The loss function $L_{\rm total}$ is defined as:

$$L_{\rm total} = w_{\rm ini}L_{\rm ini} + w_{\rm ode}L_{\rm ode} + w_{\rm data}L_{\rm data}. \tag{20}$$

Here, we set $w_{\rm ode} = w_{\rm data} = 1$, to ensure that the solution remains primarily driven by the ODE equations and observation data, which are used to solve for the parameters within the ODE system.

2.4 Optimization strategy

We adopt a multistage optimization strategy to improve training stability and accuracy for the forward problem. First, we pretrain the network using the Adam optimizer (D. Kinga & J.B. Adam) for approximately $N_{\rm Adam} \approx 8000$ iterations, which provides robust progress early in training through stochastic, adaptive-gradient updates. Subsequently, we switch to the quasi-Newton limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) optimizer (D.C. Liu & J. Nocedal 1989) for fine-tuning to refine the solution and reduce the loss more tightly once the parameters are in a favourable basin. This Adam → L-BFGS framework is commonly used in PINNs (Q. Lou *et al.* 2021), as Adam helps avoid poor local minima and L-BFGS typically improves final convergence for deterministic physics-based losses. In addition, we update the residual-balancing weights during training to mitigate scale differences among the ODE residual terms, which helps prevent any single equation from dominating the optimization. Adam facilitates rapid exploration of the parameter space with adaptive gradients, while L-BFGS achieves local precise optimization through quasi-Newton methods. The combination of the global exploration capability of Adam and the local precision of L-BFGS, along with the adaptive weight balancing constraints from the ODE equations, ensures physical consistency and numerical stability, leading to efficient convergence.

3 RESULTS

3.1 Forward problem

We apply the Runge–Kutta (RK) solver and the trained PINN to solve for the state vector ζ_2. The state vector ζ_1 is subsequently calculated from ζ_2 using eq. (11). The time variable t is defined in the interval [0, 6], corresponding to the physical time $t_{\rm real} = tL/v_*$, where L is the characteristic length and v_* is the reference velocity. The sampling interval is set to $\Delta t = 0.1$ to ensure high-resolution capture of the system dynamics. Other relevant parameters are provided in Table 1. We first run the coupled RSF model through a warm-up stage and use a point from the stabilized trajectory as the initial condition for the PINN calculation.

Fig. 3 summarizes the forward-simulation results for a transient slow-slip episode in the coupled friction–fluid system. Fig. 3(a) compares the PINN predictions with the RK reference for the components of ζ_2 (and the derived variables ζ_1), showing close agreement over the full time window. Fig. 3(b) reports the pointwise error relative to RK.

The slip rate v exhibits strongly nonlinear dynamics characteristic of RSF-governed transients: It accelerates rapidly to a peak early in the simulation and then decays gradually. The PINN captures both the onset and relaxation phases, including the sharp early-time acceleration. The shear stress $\tau_{\rm f}$ varies more smoothly and does not peak exactly at the same time as v, reflecting the interplay between the direct velocity effect (a) and the evolving state contribution (b) in the friction law. The PINN reproduces this phase shift and maintains $\tau_{\rm f}$ within the expected range throughout the transient.

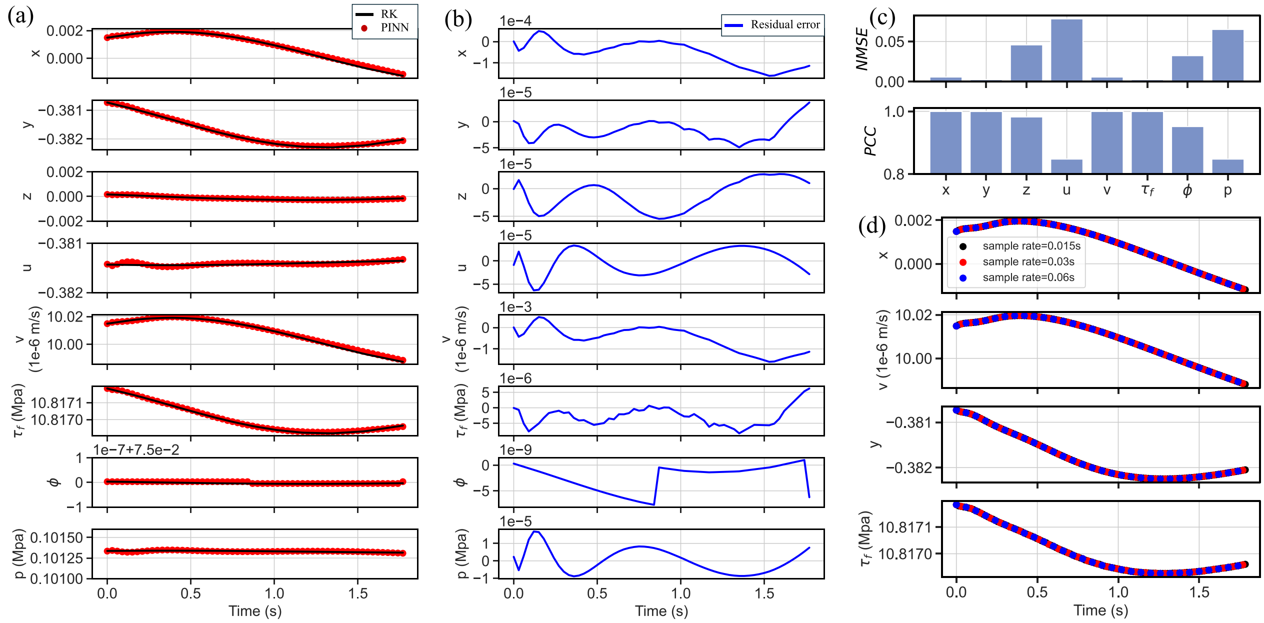

Figure 3. The solution results and residuals for the state variables ζ_2 and ζ_1. (a) Solution results from the RK method and the PINN method. (b) Residual values of the solutions from the RK method and the PINN method. (c) Quantitative accuracy measured by NMSE and PCC. (d) The solution results using different uniformly spaced sampling rates.

The hydraulic variables show coupled, time-dependent responses. Pore pressure p displays transient fluctuations during the slip episode, consistent with the governing coupling between slip, porosity evolution and fluid pressure in the model equations. Changes in p modify effective normal stress and therefore feedback on frictional strength, closing the hydromechanical loop. Porosity ϕ varies over a smaller range but exhibits a discernible change in slope during the transient, indicating a short-timescale adjustment driven by the coupled evolution equations.

To quantify agreement, Fig. 3(c) reports the Pearson correlation coefficient (PCC) and the normalized mean squared error (NMSE) between the PINN and RK solutions. Across the variables shown, the PCC exceeds 0.8 and the NMSE remains below 0.1, indicating that the PINN reproduces the RK reference with good accuracy for this transient regime. Finally, we evaluate sensitivity to sampling resolution. Fig. 3(d) shows that a network trained with $\Delta t = 0.1$ yields accurate predictions when evaluated at finer or coarser output sampling, demonstrating that the PINN represents a continuous solution over the domain rather than a discrete step-by-step integration. In contrast to RK, which advances sequentially and must be rerun if the time-step strategy is changed, the PINN can be queried at arbitrary time points after training, enabling variable-resolution output without recomputing the trajectory.

3.2 Inverse problem

PINN introduces physical constraints into neural network training. It can estimate unknown physical quantities with limited observed data. We chose σ_{n0} and c^* because they represent two distinct aspects of fault mechanics—mechanical strength and hydraulic diffusivity—which jointly control the transition from aseismic to seismic slip. The PINN framework can be used to infer the target parameters, and we need to introduce an additional data loss function (see Section 2.3). We use the shear stress, solved by the RK method, as the observed value. During the inversion process, we exclusively use the L-BFGS optimizer. This is because inversion focuses on optimization accuracy, and the quasi-Newton nature of L-BFGS is particularly well-suited for the smooth, low-dimensional parameter landscapes typical of physical models constrained by prior equations.

3.2.1 Synthetic data inversion

In laboratory earthquake experiments, shear stress can be directly obtained by dividing the shear force by the shear area. We first use shear stress τ^{obs} to invert for the normal stress σ_{n0} as an example to illustrate how PINNs can be applied to inverse problems. σ_{n0} acts as a dynamically changing variable during the PINN network optimization process. To ensure the correctness of the optimization process, other variables related to σ_{n0} need to be updated during the iterative optimization. The relationship between these variables is given by the following equations

$$\tau_0 = \sigma_{\mathrm{n0}}^{\mathrm{NN}} \mu, \tag{21a}$$

$$\kappa = \frac{kL}{a\sigma_{n0}^{NN}}, \tag{21b}$$
$$\nu = \frac{\eta v_*}{a\sigma_{n0}^{NN}}, \tag{21c}$$
$$\beta_2 = \frac{-\epsilon}{\lambda\beta\sigma_{n0}^{NN}}, \tag{21d}$$
$$\alpha = \frac{Lc^* p_\infty}{v_*\lambda\sigma_{n0}^{NN}}, \tag{21e}$$

where σ_{n0}^{NN} is the normal stress value optimized in each iteration of the PINN network. Other variables are consistent with those in Section 2.1.

When extending the PINN inversion to the hydraulic diffusivity c^*, it is crucial to note that, as the iterations progress, variables related to the inverted parameter c^* must be dynamically updated within the network

$$\alpha = \frac{Lc^{*NN} p_\infty}{v_*\lambda\sigma_{n0}}, \tag{22a}$$
$$\gamma = \frac{Lc^{*NN}}{v_*}, \tag{22b}$$

where c^*NN is the hydraulic diffusivity value optimized in each iteration of the PINN network. Other variables are consistent with those in Section 2.1. Relative to the forward problem, the inverse problem requires an additional data-misfit term in the loss function. When shear stress is used as the observational data for parameter inversion, the dimensionless network output y^{NN} must be transformed back into physical shear stress τ^{NN}:

$$\tau^{NN} = a\, y^{NN}\, \sigma_{n0} + \tau_0. \tag{23}$$

Here, y^{NN} represents the dimensionless variable y learned by the PINN at each training iteration. Based on this transformation, the loss associated with the physical observational constraint can be written as:

$$L_{data} = \frac{1}{N}\sum_{i=1}^{N}\left(\tau^{NN} - \tau^{obs}\right)^2. \tag{24}$$

Fig. 4 illustrates the optimization process and results for the inversion of σ_{n0} and c^*. As training progresses, the network gradually learns the physical laws extracted from the data, combining the observed data with the physical model to form a self-consistent and physically reasonable solution. In the optimal case, the relative error between the inverted parameter values and the theoretical values is less than 1 per cent. Physically, this demonstrates that the PINN constrained two distinct, coupled mechanisms—the mechanical strength and the fluid transport rate—using only the transient stress. Moreover, the PINN architecture inherently handles the challenges posed by sparse or noisy observed data by using the physics-based loss function to regularize the solution. We will discuss the quantitative robustness tests in Section 4.2.

3.2.2 Laboratory data inversion

We use real laboratory data to evaluate the proposed inversion framework (D. Mele Veedu *et al.* 2020). The experiments employ a double direct shear configuration, in which a granular quartz gouge layer is sheared between rough forcing blocks under a constant normal stress of 17.003×10^6 Pa. After an initial warm up stage, both shear stress and slip velocity exhibit quasi-periodic behaviour. The shear stress shows repeated loading and sudden drops, while the slip velocity displays intermittent spikes corresponding to SSEs (Fig. 5). These characteristics are consistent with synthetic slow slip cycles and provide real test data for evaluating the inversion performance of the PINN model.

Figs 6(a)–(d) show the inversion results obtained from laboratory data. Compared with the synthetic-data case (Fig. 4), the loss functions exhibit a similar overall convergence pattern during optimization, but show stronger fluctuations at the early training stage. The inverted normal stress σ_{n0} rapidly converges to a stable value, with a relative error of less than 1 per cent, indicating that the PINN framework remains robust when applied to real experimental data. For the hydraulic diffusivity c^*, we used $c^* = 10\,\mathrm{s}^{-1}$ as a nominal reference value for evaluating the inversion results, rather than as an independently measured ground-truth value. This estimate is based on the observation that, at high normal stresses, σ_n takes approximately 0.1 s to recover to the imposed value. The inversion likewise converges to the correct order of magnitude, although its relative error ($\sim$1.4 per cent) is slightly higher than that obtained from synthetic data. These results show that the proposed PINN framework can recover key physical parameters from the laboratory data. The consistency between the synthetic and laboratory inversions further indicates that the governing physical constraints help stabilize the solution.

Although we can measure shear stress directly in laboratory earthquake experiments, we cannot observe it in natural fault systems. However, we can estimate fault slip velocity from geophysical observations. We therefore test whether slip velocity can constrain the inversion of normal stress and hydraulic diffusivity. When we use shear stress as the observational data, we transform the model output into the physical observation space. However, this strategy does not work well for slip velocity. Because slip velocity in laboratory earthquake experiments is relatively small in magnitude, its direct use in the data-misfit term may weaken the contribution of observational constraints during optimization. We therefore adopt a modified constraint strategy to balance the slip-velocity observations with the

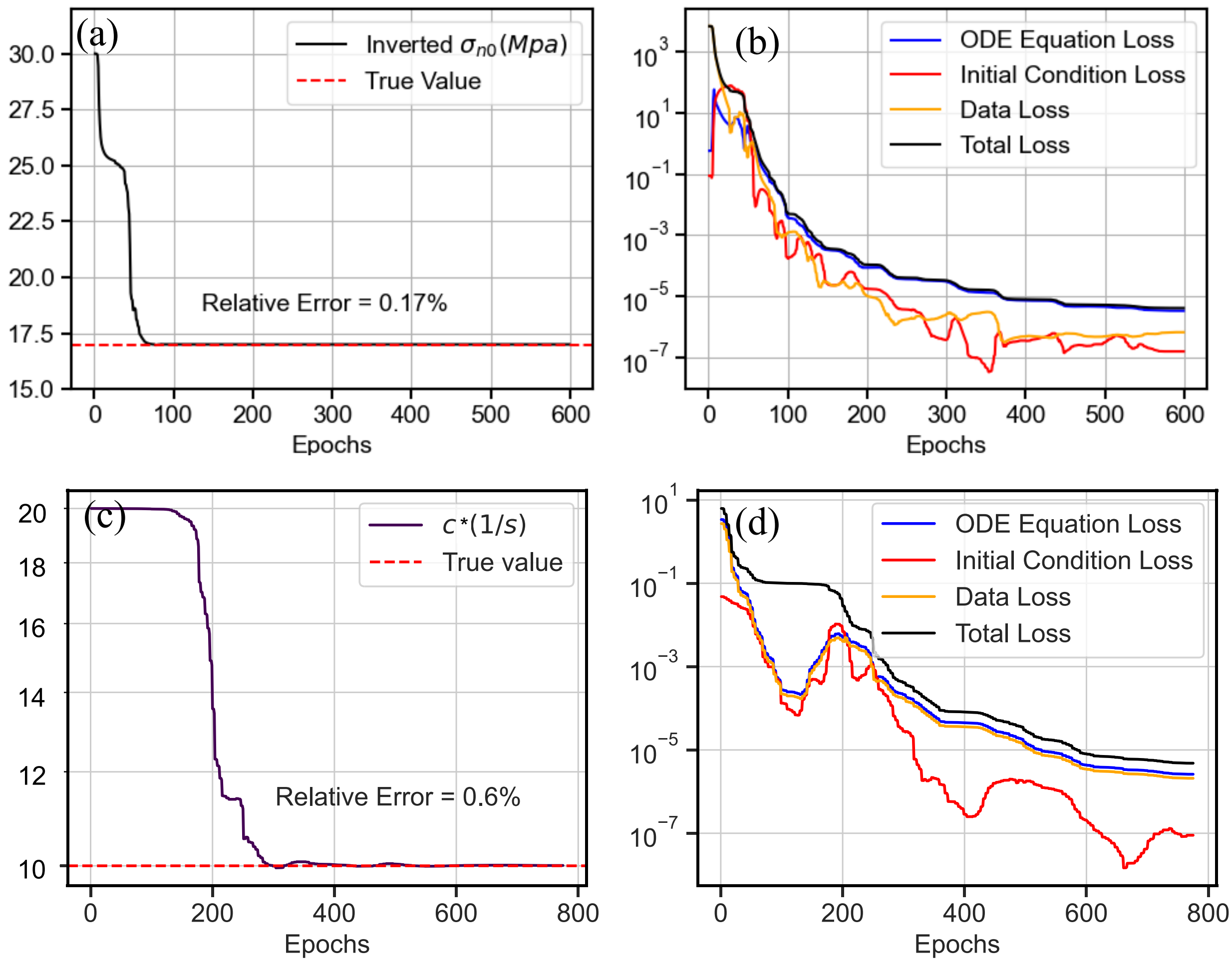

Figure 4. Inverted results of σ_{n0} and c^* using observed data. (a) Optimization process of σ_{n0} inversion using PINN. (b) Changes in each loss during the σ_{n0} inversion process. (c) Optimization process of c^* inversion using PINN. (d) Changes in each loss during the c^* inversion process.

other loss terms. We transform the observational data into the model-output space:

$$x^{\mathrm{obs}} = \ln\left(\frac{v^{\mathrm{obs}}}{v_*}\right). \tag{25}$$

We define the data loss based on the transformed slip velocity as:

$$L_{\mathrm{data}} = \frac{1}{N}\sum_{i=1}^{N}\left(x^{\mathrm{NN}} - x^{\mathrm{obs}}\right)^2. \tag{26}$$

Figs 6(e)–(h) show the inversion results using slip velocity as the observational constraint. The inversion recovers σ_{n0} with a relative error of 1.76 per cent. It recovers c^* at the correct order of magnitude, with a relative error of 4.44 per cent. The loss functions converge rapidly and remain stable. The logarithmic transformation improves the scale of the velocity constraint. Compared with the shear-stress-based inversion (Figs 6a–d), the velocity-based inversion shows slightly larger errors, especially for c^*. Nevertheless, slip velocity alone can still constrain the key physical parameters.

3.2.3 Joint inversion of multiple parameters

We show that slip velocity can invert each parameter separately. However, slip velocity reflects both the mechanical response of the fault and the evolution of pore-fluid processes. We therefore perform a joint inversion of the normal stress σ_{n0} and the hydraulic diffusivity c^*. We define σ_{n0} and c^* as trainable parameters in the network. We update them jointly through backpropagation. The optimizer acts on three sets of variables: the neural network weights, σ_{n0} and c^*. Both parameters enter the governing equations as dynamical variables:

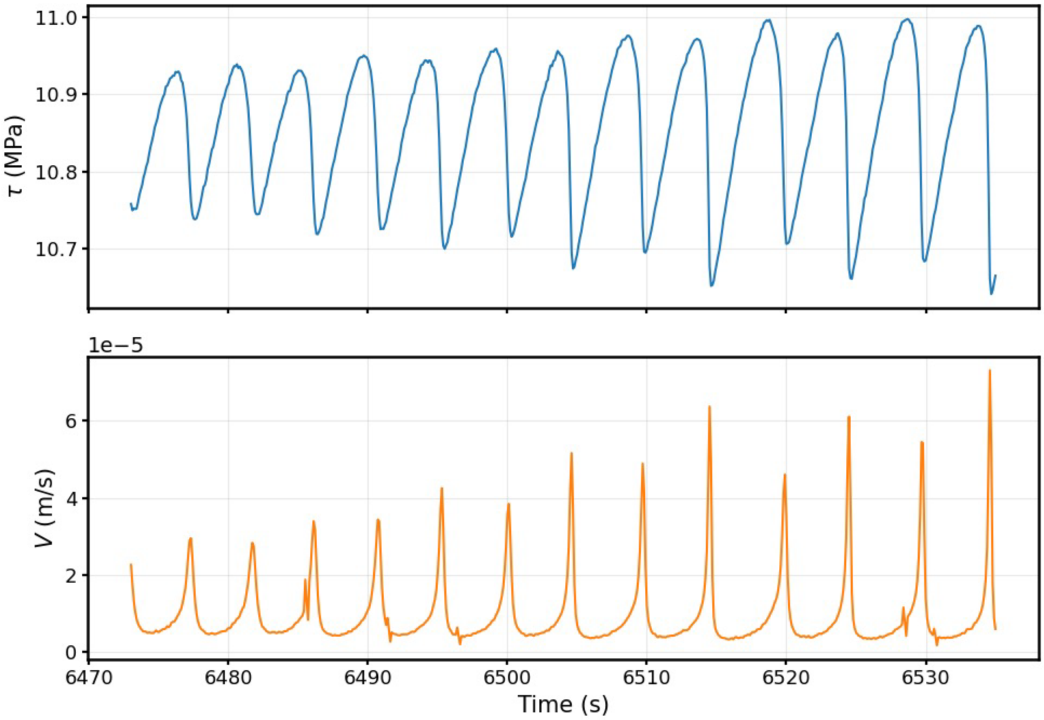

Figure 5. Laboratory fault measurements. (a) Shear stress time-series showing quasi-periodic loading and stress drops. (b) Slip velocity time-series characterized by intermittent spikes associated with episodic SSEs.

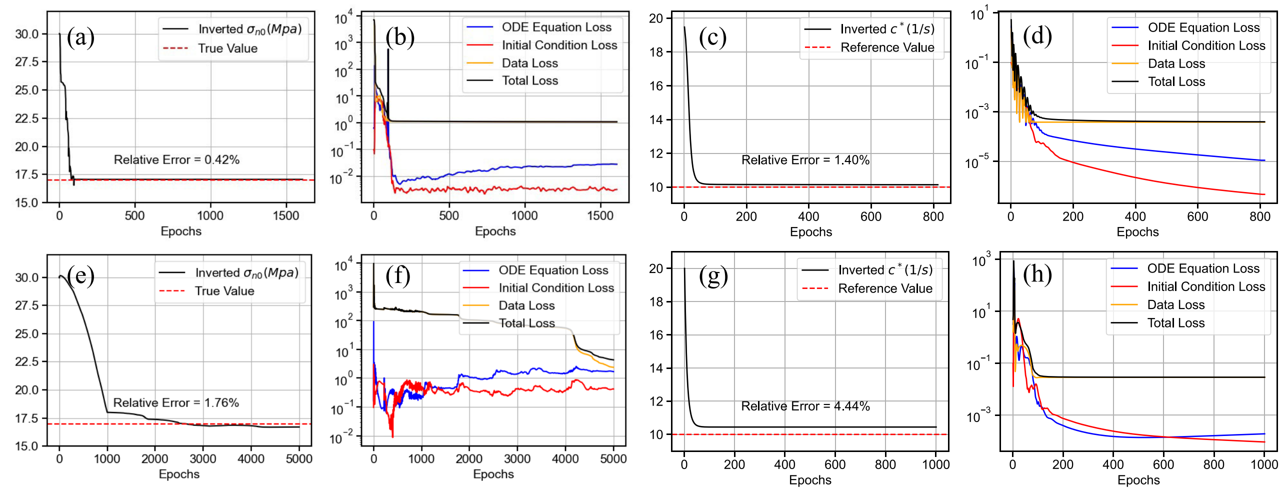

Figure 6. Inverted results of σ_{n0} and c^* using laboratory earthquake data. (a–d) Results constrained by observed shear stress. (e–h) Results constrained by observed slip velocity.

$$\kappa = \frac{kL}{a\sigma_{\mathrm{n0}}^{\mathrm{NN}}}, \tag{27a}$$
$$\nu = \frac{\eta v_*}{a\sigma_{\mathrm{n0}}^{\mathrm{NN}}}, \tag{27b}$$
$$\beta_2 = \frac{-\epsilon}{\lambda\beta\sigma_{\mathrm{n0}}^{\mathrm{NN}}}, \tag{27c}$$
$$\alpha = \frac{Lc^{*\mathrm{NN}}p_\infty}{v_*\lambda\sigma_{\mathrm{n0}}^{\mathrm{NN}}}, \tag{27d}$$
$$\gamma = \frac{Lc^{*\mathrm{NN}}}{v_*}. \tag{27e}$$

Fig. 7 shows the joint inversion results for σ_{n0} and c^* using slip velocity as the observational constraint. The joint inversion works because the two parameters leave different signatures in the time-series. The amplitude of slip or stress variations mainly constrains σ_{n}, whereas the recovery time and phase lag after each event mainly constrain c^*. For the synthetic data, the relative errors are 2.92 per cent for σ_{n0} and 2.34 per cent for c^*. For the laboratory earthquake data, the network also converges to stable solutions. The relative error of

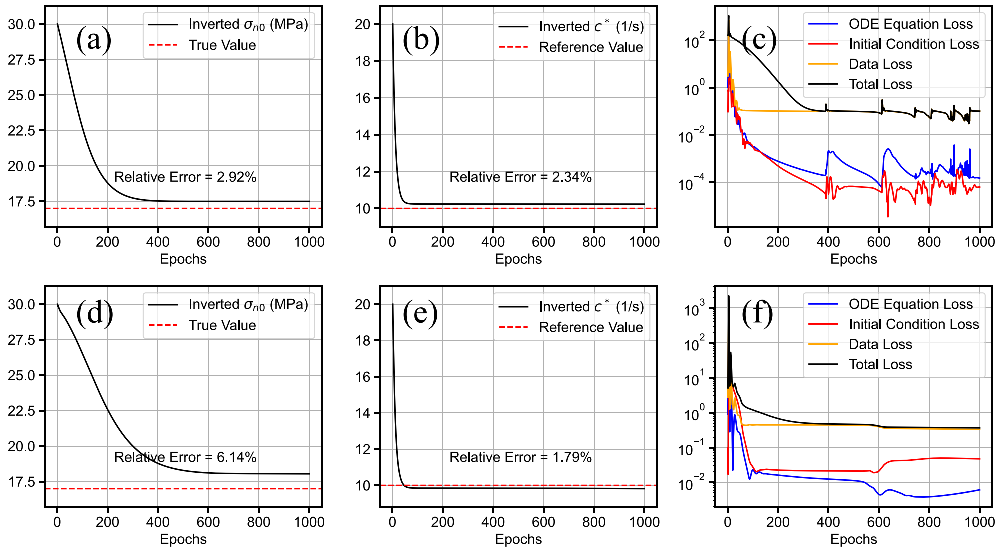

Figure 7. Joint inversion results for σ_{n0} and c^*. (a–c) Inversion results and loss evolution for the joint inversion using slip velocity from synthetic data as the observational constraint. (d–f) Inversion results and loss evolution for the joint inversion using slip velocity from laboratory earthquake data as the observational constraint.

σ_{n0} increases to 6.14 per cent, whereas the relative error of c^* is 1.79 per cent. These results show that the proposed framework can still recover the main physical parameters from real laboratory data, even when the inversion becomes more difficult. Compared with the single-parameter inversion in Fig. 6, the joint inversion shows a larger error in σ_{n0}, especially for the laboratory data. This difference reflects a stronger trade-off between parameters when the network updates σ_{n0} and c^* at the same time. This result is possible because the full time-series contains both amplitude and timing information. The amplitude of slip or stress variations mainly constrains σ_{n0}, whereas the recovery time and phase lag after each event mainly constrain c^*. Overall, joint inversion increases the complexity of the optimization problem, but the PINN framework still recovers physically reasonable parameter values and maintains stable convergence.

4 DISCUSSION

4.1 From stable sliding to slow slip

Transitions between stable sliding and episodic slow slip are a central prediction of RSF theory. A common stability measure for spring-slider systems is the stiffness ratio $S = k/k_c$, where k is the elastic stiffness of the loading system and k_c is the critical stiffness of the frictional interface. In the coupled fault-fluid setting considered here, we adopt a critical stiffness that incorporates the effects of dilatancy and hydraulic diffusion (P. Segall & J.R. Rice 1995)

$$k_c = \sigma_n \frac{b-a}{L} - \frac{-\epsilon\mu_0}{\beta L} F(c^*), \tag{28}$$

where $\sigma_n = \sigma_{n0} - p$ is the effective normal stress, σ_{n0} is the applied normal stress and p is pore pressure. $F(c^*)$ is a dimensionless function describing the stabilizing effect of dilatancy coupled with pore-fluid diffusion, with

$$F(c^*) = \left[\tfrac{1+\Lambda+\Gamma}{2} - \sqrt{\tfrac{(1+\Lambda+\Gamma)^2}{4} - \Gamma}\right], \tag{29a}$$

$$\Lambda = \tfrac{\beta\sigma_n a}{-\mu_0\epsilon}\,\tfrac{\gamma^2}{\gamma+1}, \tag{29b}$$

$$\Gamma = \tfrac{\beta\sigma_n (b-a)}{-\mu_0\epsilon}\,\tfrac{1}{\gamma+1}, \tag{29c}$$

where γ denotes the ratio of the characteristic time for state evolution to that for pore-fluid diffusion. We note that eqs (28) and (29) are valid only for the special case $L = L_\phi$. Because our laboratory simulations use $L_\phi = \rho L$, with $\rho = 0.1$, we use these equations only as a qualitative stability guide. Dilatancy stabilizes the fault through a hydromechanical feedback. As slip accelerates, the fault zone dilates and porosity increases, which lowers pore pressure when fluid diffusion cannot immediately compensate for the newly created pore space. The resulting reduction in pore pressure increases the effective normal stress, strengthens the fault and suppresses further acceleration. This stabilizing effect is represented by the negative correction term in k_c. In this framework, $S > 1$ corresponds to stable sliding, $S \approx 1$ to marginal stability (often associated with slow-slip transients), and $S < 1$ to unstable slip.

To test whether the PINN reproduces these regime changes, we keep k fixed and vary k_c by changing σ_{n0} (Fig. 8b), which modifies effective normal stress and therefore stability. As σ_{n0} increases over the range 13.50–21.98 MPa, the system approaches the critical

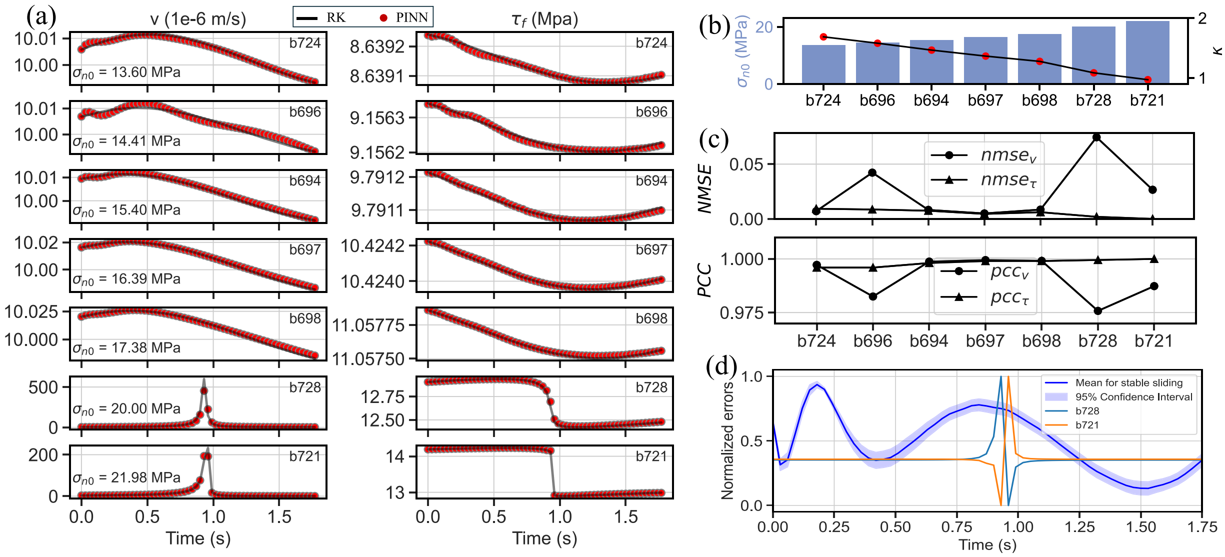

Figure 8. Results and error analysis for different normal stresses. (a) Comparison of calculation results from stable sliding to slow slip. (b) Normal stress and stiffness ratio settings. (c) NMSE and PCC of PINN and RK calculation results. (d) Variation of normalized error in calculation results over time under different scenarios.

regime and transitions from steady sliding to transient slow-slip events. The PINN captures this qualitative change in behaviour: In the stable regime, slip velocity remains low with small stress variations, whereas near the critical regime the solution exhibits intermittent transients with increased slip rate and associated shear-stress perturbations. Importantly, these transitions are recovered without explicit time stepping, since the PINN represents the solution as a continuous function of time constrained by the governing equations.

We quantify agreement with the RK reference solution using NMSE and PCC (Fig. 8c). Across the tested stress conditions, NMSE remains below 0.08 and PCC exceeds 0.975, indicating that the PINN reproduces both steady sliding and slow-slip transients with high fidelity. The error structure differs between regimes (Fig. 8d): Deviations during stable sliding are dominated by the enforcement of initial conditions, whereas in the slow-slip regime the largest errors occur during rapid transients, consistent with the difficulty of resolving sharp time-localized features with a finite set of collocation points. This highlights a practical limitation: PINN accuracy near fast transitions depends on collocation resolution and training strategy.

4.2 Effect of hydraulic diffusivity and long-term sequences

We further explore the role of nondimensional hydraulic diffusivity c^*, which controls the rate of pore-pressure redistribution in the fault zone. Varying c^* changes the coupled hydromechanical feedback and can shift the system from a steady-state attractor to oscillatory slow-slip behaviour. In our experiments, $c^* = 0.1$ produces periodic slow-slip oscillations (Fig. 9a). At $c^* = 1$, the system lies near a transitional regime and alternates between steady sliding and slow-slip episodes (Figs 9a and b).

Capturing these dynamics over long time horizons is challenging for standard PINNs because the time domain is treated globally during optimization: Increasing the simulation window increases the difficulty of minimizing the physics residuals uniformly across the domain and can degrade convergence. In our implementation, training over $t \in [0, 6]$ is sufficient to resolve short sequences but does not capture longer-term evolution that emerges over many cycles. To extend the simulated duration, we adopt a sequential time-windowing strategy (Fig. 9c). The time interval is decomposed into shorter windows; the terminal state predicted in one window is used as the initial condition for the next. This approach preserves temporal causality and reduces the optimization burden in each window. As shown in Fig. 9(b), iterating this procedure for 10 windows reconstructs long-term behaviour, including transitions between steady sliding and slow slip in the near-critical regime.

4.3 The effect of noise on the inversion results

To investigate the robustness of the PINN inversion to noise, we invert the normal stress σ_{n0} from the noisy shear stress τ_f. In realistic laboratory settings, τ_f is affected by high-frequency noise fluctuations due to the intrinsically imprecise nature of the measurements and the servo-control system. The perturbations on τ_f will affect the variable y, and we can describe the impact of noise on y using a Wiener process W_T with intensity ε_y (A. Gualandi *et al.* 2023)

$$y = \frac{\tau_f - \tau_0}{a\sigma_{n0}} + \varepsilon_y W_T, \tag{30}$$

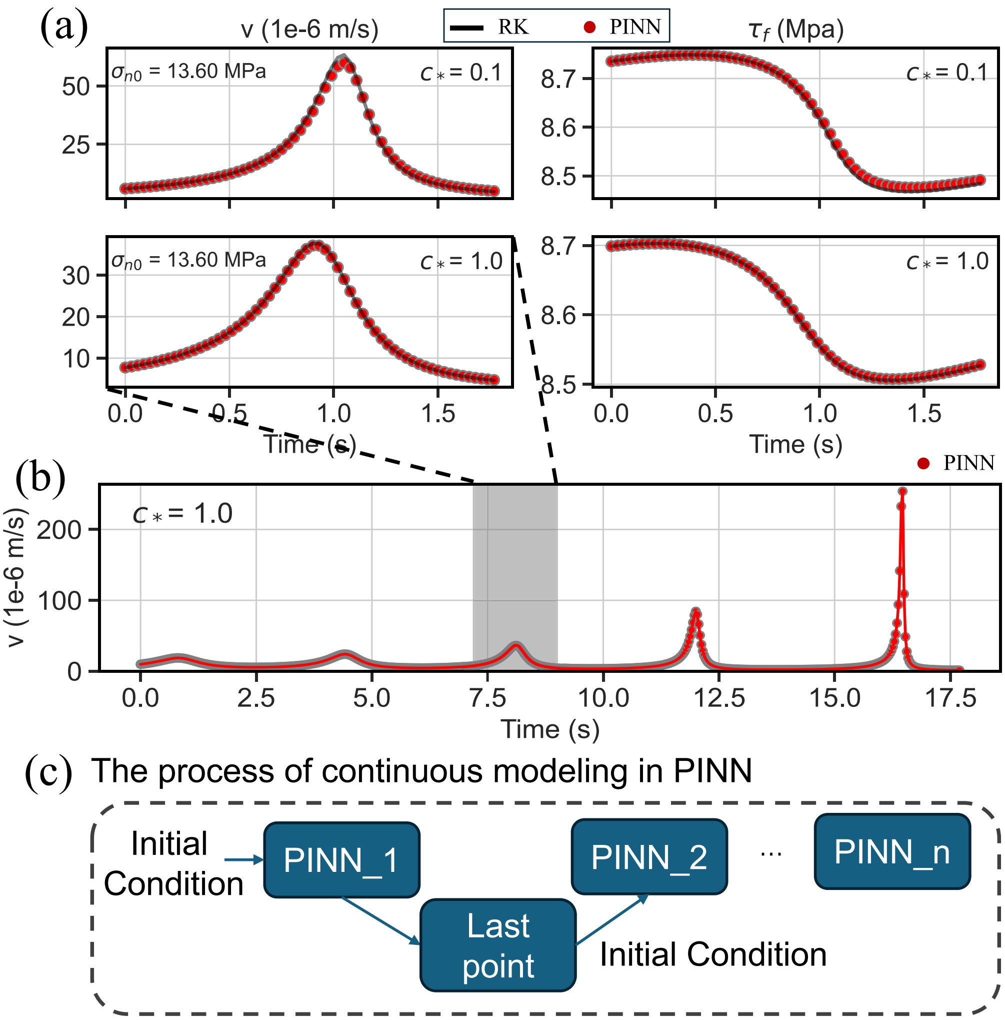

Figure 9. Simulation results for varying non-dimensional hydraulic diffusivity c^* and the sequential time-windowing strategy. (a) Detailed comparison of PINN predictions (dots) and RK reference solutions (lines) for slip rate v and shear stress τ_f. (b) Long-term evolution of slip rate v for $c^* = 1.0$. The shaded region corresponds to the zoom-in window shown in the bottom row of panel (a). (c) Schematic diagram of the sequential time-windowing strategy used to extend the simulation duration.

where $\varepsilon_y = q\varepsilon_{\tau_f}/(a\sigma_{n0})$, and ε_{τ_f} is the standard deviation of τ_f. The factor $q \in [0, 1]$ can be used to modulate the intensity of the Wiener process. After obtaining the y values at different noise levels from eq. (30), we calculate the corresponding τ_f as the simulated observations using eq. (11). This procedure yields controlled noise levels in the observation space while remaining consistent with the model variables used in the PINN loss.

We consider two frictional regimes: steady sliding (cases b697 and b698) and slow-slip events (cases b728 and b721). For each case, we evaluate three noise intensities ($q = 0$, 0.5 and 1, corresponding to noise-free, moderate-noise and high-noise conditions). In all inversions, σ_{n0} is initialized to 25 MPa. Because neural network training is sensitive to initialization, we repeat each inversion with 10 random initializations and analyse the resulting distribution of inferred σ_{n0} values (Fig. 10).

Fig. 10 shows that increasing noise primarily degrades the precision of the inversion rather than shifting the central estimate. As q increases, the kernel density estimates become broader, indicating increased variability among repeated inversions. For $q = 0$ and $q = 0.5$, the inferred means and spreads remain similar, suggesting that the inversion remains stable under low-to-moderate noise. At the highest noise level ($q = 1$), the dispersion increases substantially; for example, in the slow-slip case b721, the inferred σ_{n0} distribution broadens from 22.19 ± 0.09 MPa to 22.33 ± 0.22 MPa. Across the cases considered, the mean inferred σ_{n0} remains largely consistent, while the variance increases with noise level. These results indicate that the PINN inversion is robust in the mean estimate of σ_{n0} but becomes less precise as observational noise increases, particularly in the high-noise regime.

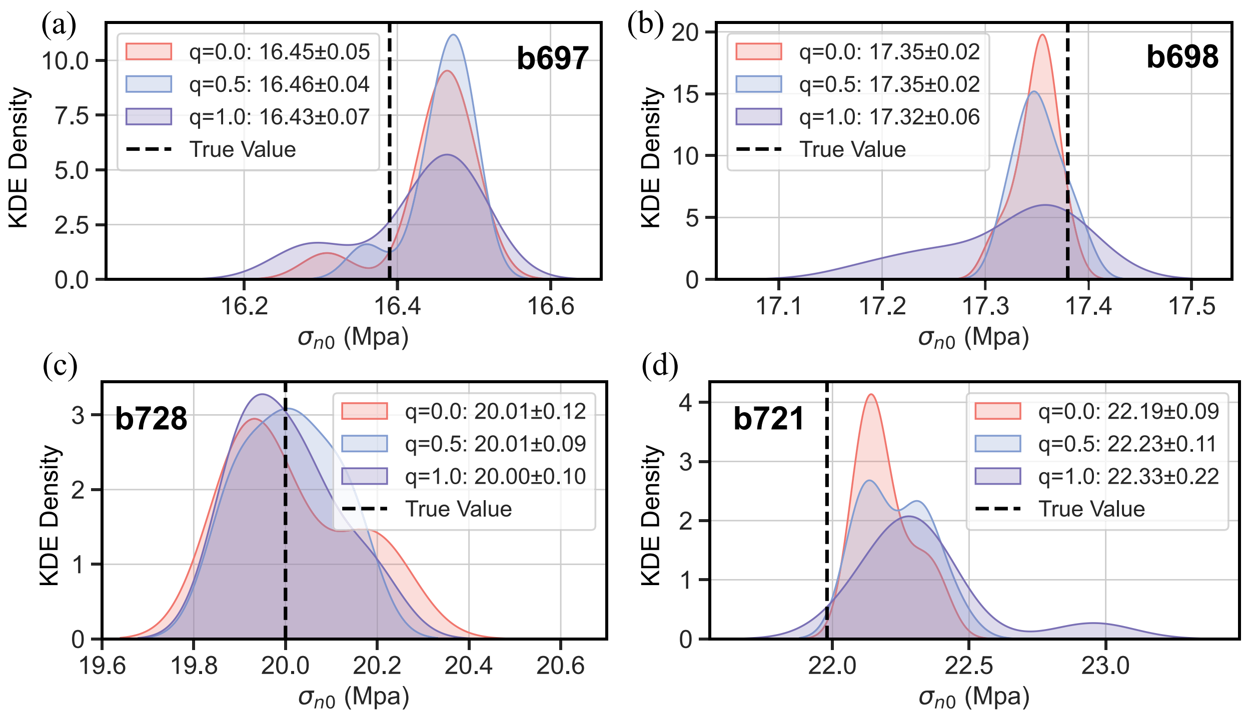

Figure 10. Kernel density estimates of the empirical distributions of the final σ_{n0} estimates obtained from 10 repeated deterministic PINN inversions with different random initializations. The curves visualize the spread of deterministic inversion results.

4.4 Optimization and loss

Training a PINN involves balancing multiple loss terms (e.g. initial conditions, data misfit and residuals from several coupled governing equations). In coupled hydromechanical systems, the residuals can differ substantially in magnitude and an unweighted loss may cause optimization to be dominated by the largest residual terms. This can slow convergence and lead to solutions that satisfy some equations more accurately than others. To reduce this imbalance, we use an inverse residual weighting strategy that rescales the contributions of the individual equation residuals during training. As shown in Figs 11(a) and (b), the residual magnitudes become comparable early in training (within the first ∼40 epochs), indicating that the reweighting stabilizes optimization and promotes simultaneous enforcement of all governing equations.

We also employ a two-stage optimizer schedule (Adam followed by L-BFGS), which is commonly used in PINN applications(Q. Lou *et al.* 2021). Adam provides robust initial progress during early training by adapting learning rates across parameters, while L-BFGS is effective for refining solutions once the optimization has entered a favourable basin. In our experiments (Figs 11c and d), Adam decreases the loss rapidly but tends to plateau or fluctuate near the minimum, whereas switching to L-BFGS yields a more pronounced reduction of the final loss and improves convergence of the physics residuals. We also observe that cases exhibiting slow-slip transients (e.g. b728) generally require more iterations than stable sliding (e.g. b694). This difference is consistent with the increased temporal complexity of slow-slip episodes, which include sharper transients and stronger nonlinearity. Because PINNs often learn smooth, low-frequency components of the solution more readily than sharp time-localized features, additional training effort and denser collocation near transients may be required to achieve comparable accuracy in the slow-slip regime(S. Wang *et al.* 2021).

4.5 Limitations of study and outlook

For simple forward simulations with known parameters and initial conditions, RK4 remains more efficient and straightforward than PINNs. Here, the forward simulations mainly serve as a consistency check against the RK4 reference solution. The main advantage of PINNs lies in inverse problems, where governing equations, observational constraints and unknown physical parameters can be combined within a unified differentiable optimization framework.

A key limitation of inverse problems in fault mechanics is that observational data are inevitably noisy, which can degrade parameter identifiability and increase uncertainty in inferred model parameters. In our noise experiments, increasing noise primarily increases the dispersion of inversion outcomes across different random initializations, indicating reduced precision in the inferred

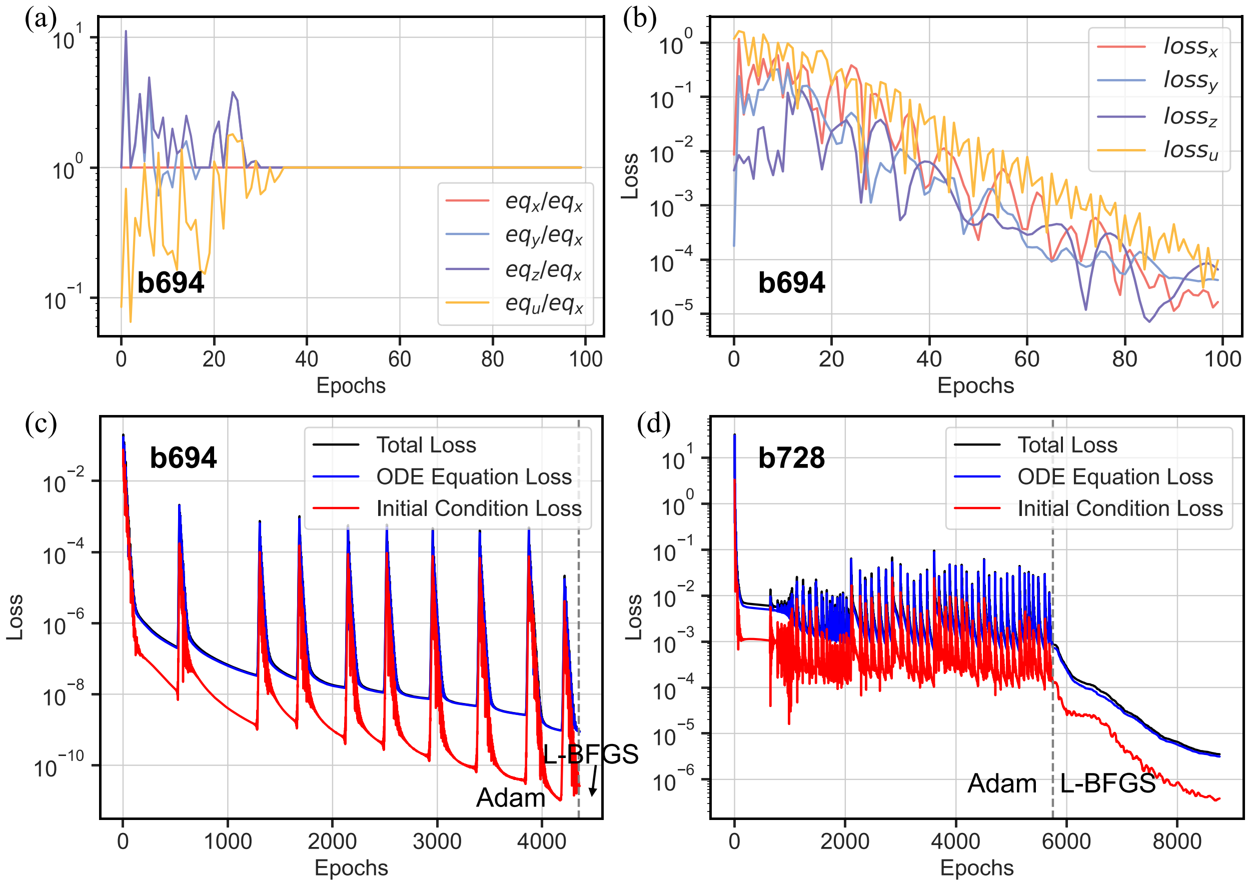

Figure 11. The optimization process of the loss for b694 and b728. (a) A comparison of the loss magnitude for the first 100 epochs of the four ODE equations in the b694 experiment. (b) The loss optimization graph for the first 100 epochs of the four ODE equations in the b694 experiment. (c) The loss optimization process for the b694 experiment. (d) The loss optimization process for the b728 experiment.

parameters. While repeating inversions with multiple initializations provides a pragmatic way to characterize variability, it is computationally expensive and does not provide a principled uncertainty estimate. A natural extension is to adopt probabilistic formulations, such as Bayesian PINNs (L. Yang *et al.* 2021), which treat unknown parameters (and potentially model discrepancy) as random variables. This approach can produce posterior distributions that quantify uncertainty and improve interpretability when data quality deteriorates.

The second limitation concerns long time horizons and multiscale dynamics. Standard PINNs optimize the solution over the entire time domain at once, and training can become increasingly difficult as the time window expands, particularly in regimes with intermittent or oscillatory transients. In this study, we partially mitigate this issue using sequential time-windowing; however, scaling to longer and more complex seismic sequences will likely require methods that better represent multiscale temporal structure. Operator-learning approaches provide a promising direction. For example, Fourier Neural Operators learn mappings between function spaces rather than a single trajectory, and can be efficient for repeated evaluations once trained (Z. Li *et al.* 2020). Exploring operator-learning or hybrid PINN–operator frameworks may therefore be beneficial for extending AI-based solvers to long-duration fault simulations.

Finally, although our model couples RSF with porosity and pore-pressure evolution, it is primarily evaluated in a laboratory earthquake setting, whereas natural faults involve broader parameter ranges, larger spatial scales and additional multiphysics processes. Three extensions are therefore especially important. First, the framework should be tested in upscaled simulations using parameter regimes representative of natural faults. Secondly, it should be extended to spatially continuous 1-D/2-D fault models, which is important for demonstrating the applicability of PINNs to real-world problems. Thirdly, the current constitutive description could be expanded to capture changes in frictional behaviour during rapid acceleration, for example, through cut-off-rate formulations (B. Shibazaki & Y. Iio 2003) and to incorporate more explicit microphysical descriptions of granular flow and state evolution, such as the Chen–Niemeijer–Spiers framework (A. Niemeijer & C. Spiers 2007). Progress along these lines, together with validation against laboratory and real observations, will be essential for translating the present proof-of-concept framework towards realistic multiphysics earthquake-cycle modelling and real-world applications.

5 CONCLUSION

We develop a PINN solver for a coupled spring-slider fault model that combines rate-and-state friction with pore-pressure and porosity evolution. The PINN represents the state variables as continuous functions of time and is trained by enforcing the governing ODE system together with initial conditions (and observational constraints for inversion). To improve training stability in this stiff, multiequation setting, we employed adaptive residual weighting and a two-stage Adam → L-BFGS optimization strategy.

In forward simulations, the PINN reproduces both steady sliding and transient slow-slip dynamics and agrees closely with a Runge–Kutta reference solution (NMSE < 0.08 across the cases tested). In inverse experiments, the framework successfully recovers key parameters (e.g. applied normal stress) and remains stable under low-to-moderate noise, with uncertainty increasing as noise amplitude grows. These results indicate that PINNs provide a viable complementary tool for solving and inverting coupled hydromechanical RSF models, particularly when a differentiable surrogate and unified forward–inverse workflow are desirable. Future work should extend the approach to longer time horizons and richer multiphysics formulations and incorporate uncertainty quantification for high-noise observations.

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SUPPORTING INFORMATION

Supplementary data are available at *GJIRAS* online.

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DATA AVAILABILITY

The model is implemented using PyTorch (A. Paszke *et al.* 2019). The available code of The PINN model can be accessed at Zenodo (J. Wang 2026). The friction experiment data used in this study are available from the Open Science Framework (D.M. Veedu 2020).

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